

# Aayush Kumar Singh

Full Stack Developer | ML Engineer

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## Professional Summary

Full Stack Developer with internship experience building scalable web applications using React.js, Next.js, Node.js, Express.js, and FastAPI. Proficient in TypeScript, REST API design, and performance optimization. Experienced with AI-integrated systems using RAG pipelines, LLM integration, and vector databases. Hackathon runner-up.

## Technical Skills

**Languages:** TypeScript, JavaScript, Python

**Frontend:** React.js, Next.js, Tailwind CSS, Bootstrap

**Backend:** Node.js, Express.js, FastAPI

**Databases:** MongoDB, PostgreSQL, MySQL, Vector Databases (ChromaDB)

**AI/ML:** NLP, Retrieval-Augmented Generation (RAG), LLM Integration, scikit-learn, MobileNet, Semantic Search

**Tools:** Git, GitHub, Vercel, Render, Netlify, Docker, Inngest

**Concepts:** REST API Design, JWT Authentication, Redux, Context API, SEO Optimization, Agile/Scrum, API Documentation

## Experience

### Web & Mobile Application Development Intern (AI Integration)

Jan 2026 – Apr 2026

Nexalaris Tech Pvt. Ltd.

- Led Team Sphinx (4 interns) delivering AI-integrated software in Agile/Scrum sprints with daily standups and stakeholder communication.
- Built *Himalayan Bodhi Spices* e-commerce platform with Next.js, TypeScript, Tailwind CSS, and Framer Motion, reducing layout inconsistencies across 10+ pages and improving mobile responsiveness.
- Designed REST API endpoints with inline documentation; performed UI, functional, and regression testing on *Subhyatra* mobile app (React Native), logging 20+ bugs across 3 release cycles.

### Frontend Developer Intern

Nov 2024 – Dec 2024

BlueFox Pvt. Ltd.

- Built a full-stack web application from scratch (React, Node.js, Express, MongoDB); improved page load performance by **40%** via code splitting and query optimization.
- Designed and documented REST APIs for a 3-developer team; managed codebase via Git branching and code reviews.

## Projects

### Employee Management System

Live

React, Node.js, Express.js, MongoDB, Inngest, JWT, TypeScript, Tailwind CSS

- Full-stack HR platform covering attendance, leave workflows, and automated payslip generation for 50+ employee profiles; reduced manual HR effort by an estimated **60%**.
- Role-based JWT auth across 2 dashboards; 15+ RESTful API endpoints with clean frontend/backend separation; Inngest automation for alerts and email delivery via Nodemailer.

### SmartPrep AI – A Virtual Teaching Assistant

GitHub

Next.js, FastAPI, MongoDB, ChromaDB, RAG, LLM, NLP, Cloudinary

- Built a full-stack learning platform with a Next.js frontend, FastAPI backend, and MongoDB — supporting PDF uploads, user sessions, multi-document ingestion, automated assessments, and progress tracking with full OpenAPI/Swagger documentation.
- Integrated RAG pipelines with ChromaDB vector search and LLM to convert uploaded PDFs into summaries, lessons, and MCQs, reducing manual study preparation time by **70%**.

## Achievements

- **Runner-up – Janakpur Hackathon 2.0** (Dec 2025) — Built a civic-tech platform with an AI legal chatbot trained on Nepal Constitution 2072 using RAG, real-time notifications, and an NGO categorization system for incident response.

## Education

### Bachelor in Computer Engineering

2022 – 2026

Institute of Engineering, Purwanchal Campus